Stablecoin StressBench, Cross-Mechanism Transfer Learning for Executable Arbitrage Under a $12\times$ Optical Gap

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Abstract

On 10 March 2023, Silicon Valley Bank's seizure forced USDC off its dollar peg. Price charts showed 34.3% of one-minute windows as profitable stablecoin arbitrage. Fewer than 3 in 100 (2.88%) survived VWAP book-walking, fees, and settlement, a 12× **gap** between visible and executable profit. A model scoring well on price labels can still lose money on every trade. We study when a learned trading model transfers across financial crises, using stablecoin depegs as the testbed. The central obstacle is distributional. A model trained on calm or single-regime data cannot identify the executable minority in a stress regime it has never seen, and every calm-trained family we test (tabular classifiers, sequence models, reinforcement learning, and changepoint detectors) fails economically despite high AUROC. The highest-AUROC model (GRU, AUROC 0.80) is simultaneously the largest money-loser (−239 bps), a direct quantification of the AUROC–P&L inversion under class imbalance. Our main result is that *cross-mechanism transfer* solves this. A meta-labeling classifier trained on an algorithmic collapse (Terra/LUNA) and applied unchanged to a bank-reserve shock (USDC/SVB) recovers +82.5 bps (51.0% of a hindsight oracle), apparently because depth withdrawal and spread widening behave alike regardless of the trigger, the order book responds to uncertainty, not its cause. *Scope.* This real cross-mechanism transfer is demonstrated for a single source → destination pair (Terra/LUNA → Tier-A USDC/SVB). Additional multi-mechanism conditions use synthetic stress events that illustrate, rather than independently confirm, the invariance hypothesis, so a second execution-grade event is the primary open validation requirement. We further isolate *supervision format* as the binding constraint. At identical positive-label density, explicit binary labels earn +82.5 bps while reward optimisation (PPO-GRU) earns −29.2 bps. To support these findings we release Stablecoin StressBench, an 18-event catalogue across seven mechanism classes, per-minute VWAP (volume-weighted average price) book-walking labels with route provenance, and a hindsight oracle ceiling, under CC-BY 4.0.

Keywords

transfer learning, meta-labeling, reinforcement learning, execution-aware labels, market microstructure, stablecoin arbitrage, financial benchmarks

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2026.

1 Introduction

Stablecoin stress is not one phenomenon but seven. From 2020 to 2023 we catalogue 18 distinct depeg episodes spanning algorithmic death spirals (Terra/LUNA, IRON/TITAN), fiat-reserve shocks (USDC/SVB), regulatory winddowns (BUSD), exchange-credit collapses (FTX, Celsius), DeFi pool imbalances (Curve/USDT), collateral liquidations (DAI Black Thursday), and real-world-asset failures (USDR, Acala). Yet every prior study examines a single mechanism, and none asks the question a trader actually faces. Of the price dislocations that appear during stress, how many can be captured at a profit?

The answer is far fewer than the chart suggests. When USDC slipped below \$1 during the March 2023 SVB crisis, Bitcoin bought with USDC looked cheaper than the same Bitcoin priced in real dollars, free money on a price chart. In practice it rarely was. By the time an order walked the book, the discount had narrowed. Taker fees and settlement delay consumed the rest. Only 2.88% of one-minute windows survived all three costs, against 34.3% that appeared dislocated, a 12× gap between visible and executable arbitrage.

We separate three layers of the same event. **Layer 1 (optical)** is what the price chart shows. A stablecoin's *basis*, its deviation from \$1, appears positive, suggesting a profitable arbitrage. **Layer 2 (executable)** is what survives real execution. A market order does not fill at the quoted price. It consumes successive price levels in the order book, and the resulting VWAP (volume-weighted average price, $\bar{P} = \sum_i P_i Q_i / \sum_i Q_i$ across consumed levels) worsens with size. Layer 2 keeps only the minutes where the VWAP-filled profit, after fees and settlement delay, is still positive. **Layer 3 (captured)** is what a model can flag in advance. Gap 1 is the distance from Layer 1 to Layer 2. Gap 2 is the distance from Layer 2 to Layer 3. We call Layer-1-only dislocations ***optical*** (real on the chart, illusory in the order book) and the survivors ***executable***. The two gaps have distinct causes. Gap 1 is a measurement problem. Price labels overstate tradeable performance by a factor of twelve. Gap 2 is a training problem. No model trained on calm data can find the executable minority in a stress regime it has never seen. Figure 1 shows all three layers for the SVB test window and traces one representative false-positive minute from apparent profit to net loss. Basis points (bps) are hundredths of a percent (100 bps = 1%).

We establish each gap against a ground truth, mirroring the sanity-check-then-hard-case logic of controlled studies. Terra/LUNA, a mechanistically independent algorithmic collapse, serves as our known reference. It reproduces the optical-to-executable gap at 5.9× (2.40% executable) and supplies a clean cross-mechanism training signal. USDC/SVB, a fiat-reserve shock with route-complete order-book depth, is the execution-grade test case, where the gap reaches 12× (2.88% executable) and widens to 14× under a 20% depth haircut. That the same gap appears in two unrelated mechanisms tells us the phenomenon is structural, not an artefact of either event. Only the Tier-A execution labels are SVB-specific.

The one approach that crosses zero exploits a simple regularity. The order book responds to uncertainty, not to its cause. Depth withdrawal and spread widening look identical in an algorithmic collapse and a bank-reserve shock, so a meta-labeler trained on Terra/LUNA recognises the executable signature in SVB without ever seeing it. This single transfer recovers half the return available to a hindsight oracle.

Contributions.

(1) **Cross-mechanism transfer for executable arbitrage.** A meta-labeling classifier trained on one crisis type (Terra/LUNA) and applied unchanged to a different, real Tier-A event (USDC/SVB) recovers +82.5 bps (51.0% oracle capture), a single demonstrated cross-mechanism transfer. The mechanism-invariance hypothesis (depth withdrawal, spread widening), which SHAP attribution supports, is further *illustrated* by synthetic multi-mechanism conditions (four-event pooled, 51.8%). We are explicit that these synthetic sources do not independently validate generality, which awaits a second execution-grade event. (§6.3)

(2) **Supervision format, not label density, as the binding constraint.** At identical 17% positive-label density, explicit binary supervision earns +82.5 bps while reward optimisation (a conditioned PPO-GRU) earns −29.2 bps, locating the difference in how the learning signal is delivered rather than in data scarcity. Every other calm-trained family (tabular, sequence, class-balanced, regression, and changepoint) also fails, ruling out architecture and features as the remedy. (§6.2, 6.3.2)

(3) **An execution-aware benchmark that makes these claims measurable.** StressBench supplies the apparatus. An 18-event source-tracked catalogue across seven mechanism classes, per-minute VWAP book-walking labels with route provenance, and a *hindsight oracle ceiling*, the profit achievable by a hypothetical agent that knows at each minute exactly which windows are executable, forming an upper bound on any model's performance. Measuring oracle capture (recovered profit as a fraction of the oracle) provides a principled denominator that raw bps figures lack. It exposes a 12× optical-to-executable gap, so a model scored on price labels overstates tradeable performance by roughly twelve times, an evaluation-integrity result for any AI trading model on this data. We further show this gap is **venue-specific**, 12× on the CEX order book but ≈ 1× on real on-chain AMM execution across all five episodes, locating the barrier in CEX microstructure, not in depegs. (§3–5, 6.1)

2 Related Work

Stablecoin stress and depeg detection. Uhlig [1] and Griffin and Shams [2] analyse Terra/LUNA and stablecoin non-fundamental demand. Hernandez Cruz et al. [21] study DEX liquidity during the SVB collapse without modelling CEX execution costs. Cintra and Holloway [9] detect the Terra depeg 12 hours early via BOCPD on AMM reserves. We replicate their approach on CEX cross-quote data and find AUROC 0.229, confirming that AMM and CEX signals differ fundamentally. Wu and Liu [20] study tail spillovers across stablecoin designs without per-minute execution analysis. None of these works provide VWAP-grade execution labels, a hindsight oracle, or cross-mechanism transfer evaluation. Our design therefore follows three principles from the broader literature. Cont and Kukanov [6] show VWAP book-walking is the correct profitability measure for sequential execution, we implement this exactly. Shleifer and Vishny [4] show that visible arbitrage gaps need not survive execution frictions, our 12× gap is a direct measurement of this effect at one-minute resolution. And Lopez de Prado [8] shows that a meta-labeler trained on the execution outcome outperforms direction predictors, our cross-mechanism result is the evidence that this extends across stress mechanisms.

Limits to arbitrage and execution frictions. Shleifer and Vishny [4], Gromb and Vayanos [5], and Makarov and Schoar [7] show that visible price gaps need not be profitable once execution frictions are included. Our 12× ratio is this effect measured at one-minute resolution. Hautsch et al. [3] quantify settlement latency as a first-order cost on blockchain assets. Cont and Kukanov [6] establish VWAP book-walking as the correct profitability measure. Gogol et al. [18] measure Maximal Arbitrage Value on AMM-to-CEX routes. Their partial execution labels cover AMM venues only and lack a hindsight oracle. Adams et al. [19] study swap costs on Uniswap without stress-event execution analysis.

ML and RL for execution. Lopez de Prado [8] introduces *meta-labeling*. A two-stage architecture in which a primary model (here, the Layer-1 dislocation flag) proposes candidate trades, and a secondary *meta-labeler* learns which of those proposals will actually be profitable. The meta-labeler thus predicts the binary label "this candidate survives execution" rather than predicting the market direction from scratch. We extend this to the *cross-mechanism* setting, where the meta-labeler is trained on one crisis type (Terra/LUNA) and applied unchanged to another (USDC/SVB). Moallemi and Wang [16] and Ning et al. [17] supply the finite-horizon MDP framing we adopt for the RL baseline. Mohl et al. [22] and Olby et al. [23] operate in reactive LOB environments with market-impact feedback. Our diagnostic uses historical replay without feedback, isolating training supervision as the variable.

3 Benchmark Design

3.1 Data Sources and Instruments

The 18-event catalogue defines the historical stress universe. The six-window empirical panel (56,134 one-minute bars) defines the model dataset, and the Tier-A SVB windows define the execution-grade testbed. Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels use three depth legs across two centralised exchanges (CEXs, i.e. Binance and Coinbase). The legs are Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference). The BTC-USDC buy leg uses kline-proxy depth for the SVB window because the perpetual contract did not exist on Binance until January 2024. A 20% depth haircut confirms robustness (gap widens to 14×, oracle stays above +130 bps).

Data quality varies by leg, and every record carries a provenance tag (Table 1).

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	partial/proxy, real L2 where archived	real L2 (futures)
Sell BTC-USDT	raw/futures depth	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024. All 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 one-minute bars).

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

3.2 Event-Based Split Design

Splits are defined by event identity, so no model trained on the calm split has encountered stress dynamics. Threshold calibration uses the Terra/LUNA validation split, an independent stress event, so the SVB test split is never observed during model development. The 2.88% positive rate in the test split implies severe class imbalance. We therefore report economic net bps as the primary criterion. AUROC and AUPRC are secondary metrics.

Five leakage checks confirm validity. Label shuffles produce negative P&L. Lookahead does not improve performance. Train and test are separated by event identity.

The full 18-event catalogue with per-event mechanism taxonomy and Tier classifications is available in the project repository.

3.3 Benchmark Positioning

Against the closest prior work, Cintra and Holloway [9], Hernandez Cruz et al. [21], Gogol et al. [18], and Adams et al. [19], no prior benchmark combines per-minute executable CEX labels, route-level L2 provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation. StressBench is the first to provide all four.

Feature sets. Four nested sets (`price_only` ⊂ `price_plus_book` ⊂ `price_book_frag` ⊂ `price_book_settle`) isolate marginal signal from price basis, microstructure (spread, depth, imbalance), fragmentation, and on-chain settlement proxies (kline-proxy for SVB/Terra windows, Tier B).

4 Execution-Aware Labels

4.1 Three-Layer Framework

Figure 1 structures the paper's central argument visually. Layer 1 (optical) contains every minute where the price chart shows a profitable gap. Layer 2 (executable) contains the minutes that remain profitable after VWAP book-walking, taker fees, and settlement latency. Layer 3 (captured) contains the minutes a deployed model can identify in advance. Gap 1 is the distance from Layer 1 to Layer 2. Gap 2 is the distance from Layer 2 to Layer 3. Both gaps are nonzero, independent, and large. Panel (b) of Figure 1 traces one representative false-positive minute through the full cost stack, showing why 34.3% optical activity collapses to 2.88% executable.

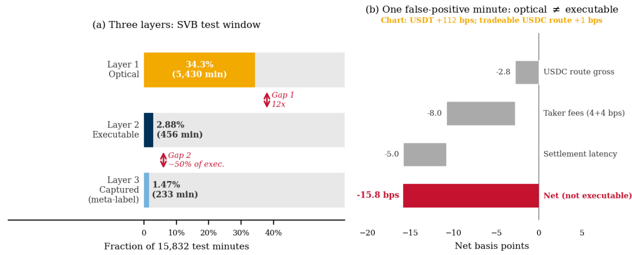

Figure 1: (a) Three-layer framework for the SVB test window (15,832 min). Layer 1 optical (34.3%), Layer 2 executable (2.88%), Layer 3 captured by cross-mechanism meta-labeling (1.47%). Gap 1 (12×) is a measurement problem. Gap 2 is a training problem. (b) Worked example. A minute that appears profitable on the chart nets −15.8 bps after VWAP book-walking, fees, and route-direction mismatch.

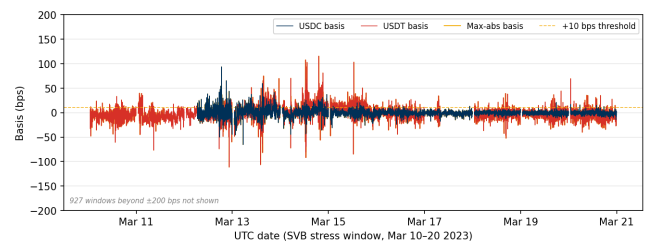

Figure 2: Cross-quote basis during the SVB test window (Mar 10–20 2023). USDC basis (navy) exceeds 10 bps in 12.4% of minutes, USDT basis (red) in 27.4%, and max-abs (gold) in 34.3%. The price-to-execution ratio is 12×.

4.2 Cross-Quote Basis and Labels

The cross-quote basis is the percentage gap between BTC priced in a stablecoin (e.g. BTC-USDC) and BTC priced in real dollars (BTC-USD on Coinbase). A non-zero basis signals an apparent dislocation. Three equations define the label construction.

$$b_{\mathrm{USDC}}(t) = 10{,}000 \times \frac{P_{\mathrm{BTC/USDC}}(t) - P_{\mathrm{BTC/USD}}(t)}{P_{\mathrm{BTC/USD}}(t)} \quad [\mathrm{bps}] \tag{1}$$

$$b_{\max}(t) = \max(|b_{\mathrm{USDC}}(t)|,\ |b_{\mathrm{USDT}}(t)|) \tag{2}$$

$$y_{\mathrm{basis}}(t) = \mathbf{1}[|b_{\mathrm{USDC}}(t+h)| > \tau] \tag{3}$$

The primary task is `basis_usdc_1m_gt10bps` (τ=10 bps, h=1 min).

4.3 VWAP Execution and Net Profit Labels

$$\mathrm{net}(q,t) = 10{,}000 \times \left[\frac{\mathrm{VWAP_{sell}} - \mathrm{VWAP_{buy}}}{P_{\mathrm{ref}}} - f_{\mathrm{buy}} - f_{\mathrm{sell}} - \delta_{\mathrm{settle}} \right] \quad [\mathrm{bps}] \tag{4}$$

$$y_{\mathrm{arb}}(q,h,t) = \mathbf{1}\left[\max_{s \in [t,\, t+h]} \mathrm{net}(q,s) > c \right] \tag{5}$$

Profit threshold c=10 bps, taker fee 4 bps per side, and δ=5 bps. The δ assumption is validated against 100K on-chain USDC ERC-20 transfers. Median cost was 1.4–3.2 bps on nine of ten SVB days. The one exception (Mar 11 gas spike) saw P95 reach 8.6 bps.

$$\mathrm{Oracle}(q) \;=\; \frac{1}{|\mathcal{T}^+|}\sum_{t\in\mathcal{T}^+}\mathrm{net}(q,t), \qquad \mathcal{T}^+ = \{t : \mathrm{net}(q,t) > c\} \tag{6}$$

The oracle is a hypothetical strategy that trades every profitable window with perfect hindsight and represents an upper bound no deployable model can exceed. Oracle capture $\kappa_M = \bar{r}_M/\mathrm{Oracle}(q)$ is the fraction of the oracle's per-trade return recovered by model M.

5 Models and Evaluation

5.1 Model Ladder

The ladder is a sequential elimination experiment with one result at every rung. Every family fails for the same reason, distributional mismatch between calm training and stress test.

Architecture. No help. The GRU posts the highest AUROC among sequence models (0.80) and the worst economic return (−239 bps, Table 6). Discriminability does not translate to profit at 2.88% positive rate.

Feature engineering. No help. All four nested feature sets produce negative returns. Depth and settlement signals offer no lift when the training split contains zero stress-regime examples.

Class balancing. No help. Cost-sensitive reweighting cannot synthesise stress dynamics from calm data. The issue is distributional, not proportional.

Regression targeting. No help. ExpNetProfitRegressor optimises the economic criterion directly and reaches −73.8 bps. Label mismatch is not the explanation.

Regime detection. No help. BOCPD, CUSUM, and EWMA all fail because CEX basis is mean-reverting, not change-point structured (BOCPD AUROC 0.229).

RL reward optimisation. No help at identical density. A conditioned PPO-GRU at 17% primary-signal density earns −29.2 bps. The reward signal is not a substitute for explicit labels even when positive-example density is matched.

The cross-mechanism extension of meta-labeling operates in three steps. Meta-labeling [8] is a two-step filter. A cheap rule flags candidate minutes, then a learned classifier keeps only the ones likely to survive execution.

(1) **Primary filter.** Flag t if $|b_{\mathrm{USDC}}(t)| > \tau_1$ (τ_1=10 bps).
(2) **Secondary classifier.** Fit f_θ on $\{(x_t,\, y_{\mathrm{arb}}(t))\}_{t\in\mathcal{E}_{\mathrm{train}}}$, where $\mathcal{E}_{\mathrm{train}}$ is a mechanistically distinct stress event.
(3) **Entry.** Signal at t iff primary fires and $f_\theta(x_t) > \theta^*$, where $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i>\theta} \mathrm{net}(q,t_i)$ on the Terra/LUNA calibration block.

5.2 RL Execution Policy Baseline

The entry-timing problem is framed as a finite-horizon MDP. At each minute t the agent observes the 30-step lookback of `price_plus_book` features, selects $a_t \in \{\texttt{enter}, \texttt{wait}\}$, and receives reward $r_t = \mathrm{net}(q,t)$ on entry, 0 on wait. We train a PPO agent with a GRU policy network (64 units, 30-step lookback) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on USDC/SVB without retraining. This directly tests whether the sequential timing framing overcomes the distributional mismatch that defeats all calm-trained models.

Table 3: Master oracle and gap table. Every oracle value cited in the paper appears here. Gap = oracle minus best deployable model.

Task / Window	N	Oracle	Best model	Gap	AUPRC
`basis_usdc_1m` (full test)	15,832	+161.7 bps	0 bps[a]	162 bps	0.253
`exec_arb_q10k_5m` (full test)	15,832	+224.6 bps	−41.1 bps	266 bps	0.060
`exec_arb_q50k_5m` (full test)	15,832	+146.8 bps	−112.7 bps	260 bps	0.063
SVB stress phase only (Mar 10–14)	7,189	+259.9 bps	–	–	–
Terra/LUNA validation split	11,526	+87.6 bps	–	–	–

[a] Best calibrated model is NoTrade (0 trades).
Kline-proxy buy leg. 20% haircut bounds in §3.1.

Table 4: Price-to-execution gap ($10K, 1-min horizon, SVB test split).

Thresh.	$P_{\max}$	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

5.3 Calibration and Evaluation

Net bps denotes mean net basis points per triggered trade after VWAP, taker fees, and settlement penalties. Thresholds are calibrated on the validation split by maximising $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i>\theta} \mathrm{net}(q,t_i)$ subject to a minimum of 25 trades. Primary metrics are mean net bps per triggered trade and oracle capture κ_M. Secondary metrics are AUROC and AUPRC. NoTrade at 0 bps is the economic floor.

Positive density clarification. Three densities appear in the paper. 2.88% is the fraction of SVB test minutes that are executable. 2.30% is the fraction of the full Terra/LUNA stream that is executable. 17% is the fraction of primary-signal minutes (those where $|b_{\mathrm{USDC}}| > 10$ bps) that are executable in Terra/LUNA. This is the density seen by the conditioned PPO-GRU and the meta-labeler secondary classifier during training. The RL-vs-meta-labeling comparison holds density fixed at the 17% figure.

Five evaluation tasks ship with StressBench v1.0, training on the calm or Terra block, validating on the Terra calibration block, and testing on SVB. The tasks are `optical_basis` (AUROC/AUPRC), `exec_arb_q10k` and `exec_arb_q50k` (net bps, oracle capture), `cross_mech` (net bps), and `policy_entry` (net bps, trades). Valid submissions must report net bps per trade, oracle capture, cost assumptions, notional size, and data tier used. Models are ranked by net bps, not AUROC.

6 Results

Table 3 maps every oracle figure in the paper to its task and window. All subsequent references point back to one of these rows.

6.1 The 12× Execution Barrier

At a 10 bps threshold, 34.3% of test minutes are optically dislocated but only 2.88% are executable after VWAP slippage, taker fees, and settlement latency. The ratio persists across all basis levels and widens with notional size. At $500K notional the executable rate collapses to 0.03%. Book depth, not fees, is the binding institutional constraint.

Block-bootstrap CIs (60-min blocks, 2,000 resamples) give an executable rate 95% CI of [1.47, 4.59]%, a ratio of [7.6, 23.0]×, and an oracle of [+213.9, +304.2] bps.

Table 5 shows the stress/recovery split. All executable windows fall in the SVB stress phase. The recovery phase contains zero executable windows despite 21.6% optical activity, confirming that depth returns before the basis closes. Terra/LUNA independently replicates the gap at 5.9× (2.40% executable of 14.0% optical), confirming that the optical-to-executable phenomenon is not specific to reserve-bank shocks. The Tier-A execution-grade labels are one event. The underlying finding is two.

If only one minute in twelve is tradeable even with perfect hindsight, the next question is whether any model can find those minutes in advance.

Table 5: Optical and executable diagnostics by window. SVB stress/recovery are Tier-A execution-grade. Terra/LUNA is validation-grade only.

Window	Min.	Optical	Exec.	Oracle
SVB Stress (Mar 10–14)	7,189	53.0%	6.63%	+259.9 bps
SVB Recovery (Mar 15–20)	8,643	21.6%	**0.00%**	–
Terra/LUNA (May 2022, val.)	11,526	14.0%	2.40%	+87.6 bps

The gap is robust across the full cost-parameter space. Sweeping taker fees from 2 to 10 bps and settlement penalty from 0 to 10 bps moves the executable rate between 2.84% and 3.66%.

The gap is venue-specific. On-chain AMM execution closes it across all five episodes. To test whether the optical-to-executable gap is intrinsic to depegs or specific to CEX order-book microstructure, we measure it on *real* on-chain pool data (Curve/StableSwap reserve state → implied price and $10k slippage, sourced on-chain) for *all five* episodes. Defining an AMM-executable window as one whose pool dislocation survives round-trip slippage and the 1 bp swap fee, the gap collapses to ≈ 1× in **every event**. 2,622/2,622 (100%) **of optically dislocated windows are capturable** at $10k notional across 4,200 clean on-chain rows (USDT/Curve, USDC/SVB, Terra/LUNA, FTX, BUSD), with median $10k slippage of only 0.8–5.5 bps even where the median dislocation ranges from sub-bp to 522 bps. The contrast with the 12× CEX gap, measured on the *same* SVB event, is the point. The optical-to-executable barrier is a property of *CEX order-book microstructure* (thin depth, taker fees, settlement delay), not of stablecoin depegs per se. On an automated market maker, where execution cost is the bounded slippage of the pool invariant, a visible dislocation is almost always a real one. This is, to our knowledge, the first multi-event measurement separating the optical-executable gap by venue.

6.2 Calm-Training Failure

The central finding is not that individual models fail, but that all six families converge to the same result. No architecture overcomes the distributional mismatch between calm training and stress test. This convergence across every model family tested is itself the diagnostic. The failure is structural, not a gap in model coverage. Table 6 shows the tabular model results on `basis_usdc_1m_gt10bps`. Oracle gap values map to Table 3.

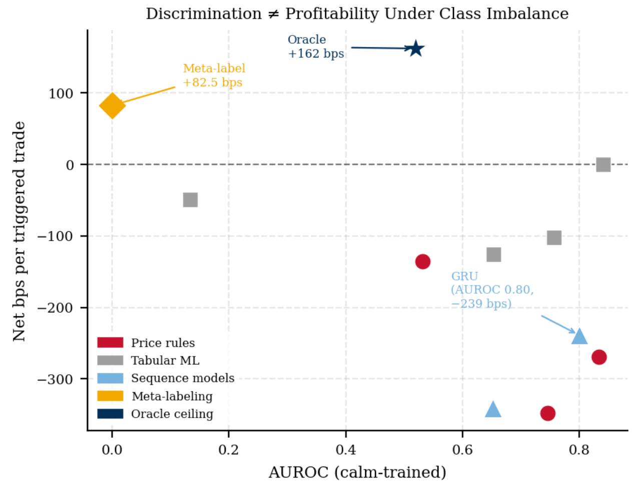

Figure 3: AUROC vs. net bps for all models on the SVB test split. Calm-trained models cluster in the upper left, high discrimination, negative P&L. Meta-labeling (orange diamond, +82.5 bps) occupies the unique profitable region with near-zero AUROC because it is activated only on transferred stress-regime depth windows, not on calm-data patterns. The GRU (AUROC 0.80, −239 bps) is annotated explicitly to illustrate the inversion. Oracle (navy star) is the achievable ceiling.

Table 6: Tabular model ladder (`basis_usdc_1m_gt10bps`). Hit% is fraction of executable trades. †Oracle ceiling. *0-trade degenerate.

Model	Feat.	Net bps	Trades	AUROC	Hit%
Oracle†	–	+161.7	316	–	100%
`lgbm`*	all	–	0	0.653	–
`logistic`	all	−49.1	1	0.133	0%
`gross_arb`	px	−136.0	611	0.531	4%
`price_10`	px	−269.5	2002	0.833	6%
`price_25`	px	−347.3	1025	0.746	5%
`no_trade`	–	0.0	0	–	–
`ExpNet_lgbm`	px	−73.8	537	–	6%
`ExpNet_xgb`	px	−57.2	70	–	9%
`lgbm_weighted`	pb	−65.9	2	0.497	0%
`xgb_weighted`	px	−90.5	30	0.631	0%
`rf_balanced`	px	−89.2	11	0.467	0%

px = `price_only`, pb = `price_plus_book`, all = `price_book_settle`.

Price-rule strategies overtrade massively (−136 to −347 bps) because false positives from route-direction mismatch dominate. $\frac{1}{3}$ fire on USDT dislocations while the USDC route is simultaneously non-executable. Sequence models (GRU, AUROC 0.80) earn −239 bps. High discrimination does not imply positive P&L when no threshold can isolate the 2.88% executable minority. Figure 3 plots AUROC against net bps for all calm-trained models. The relationship is inverted. The highest-AUROC model (GRU, 0.80) is the second worst in P&L. Meta-labeling, by contrast, carries essentially no AUROC signal on the calm-data benchmark (it fires only on transferred stress-regime windows) yet is the only approach that earns positive net bps.

Figure 4 decomposes the 260 bps oracle gap into false-positive drag (dominant), timing error (40–80 bps), and sizing error (40–100 bps).

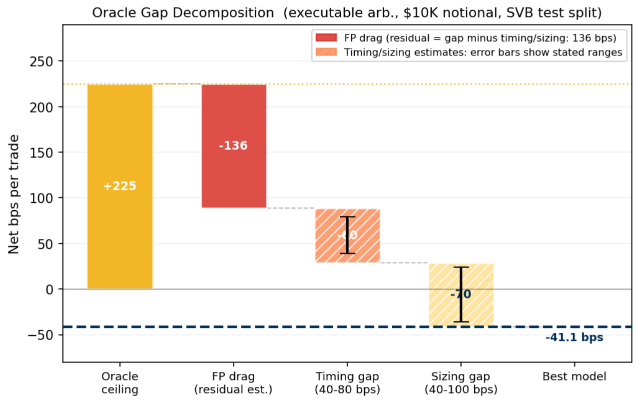

Figure 4: Oracle gap decomposition ($10K notional, SVB test split). False-positive drag is directly measured from false-positive trades. Timing error (40–80 bps) and sizing error (40–100 bps) are estimated from entry-delay and notional-scaling analysis.

6.3 Cross-Mechanism Transfer

Cross-mechanism transfer means training on one type of crisis and testing on a completely different one. Here, the filter trained on Terra/LUNA's algorithmic collapse is applied unchanged to SVB's reserve-bank shock. This approach is the only method that crosses zero, earning +82.5 bps (51.0% oracle capture on the `basis_usdc_1m` task, oracle +161.7 bps from Table 3). A reinforcement learning agent given the same 17% primary-signal density earns −29.2 bps. The 111.7 bps gap points to explicit binary labels as the ingredient that reward optimisation cannot replicate at sparse positive rates. We treat this as a hypothesis for the multi-mechanism RL experiment, not a causal claim (Section 6.3.2).

6.3.1 Cross-Mechanism Transfer and Training Diversity. Meta-labeling earns +82.5 bps on the one real cross-mechanism transfer (Terra/LUNA ⟶ SVB) and positive returns on synthetic multi-mechanism stress events (Table 7). The synthetic four-event pooled condition reaches +83.7 bps (51.8% oracle capture). As detailed below, the synthetic rows illustrate the mechanism-invariance hypothesis rather than independently confirm it.

We are deliberate about what this table does and does not establish. Terra/LUNA is the *one* transfer backed by real catalogue source data, and it recovers +82.5 bps (n = 397, 51.0% of oracle) into the real Tier-A SVB test split. The remaining single-source and pooled conditions (Celsius/3AC, FTX, BUSD) use *synthetic* stress events calibrated to the historical catalogue (§6.3). Because their generator embeds the same depth-withdrawal/spread signature in every mechanism, they *illustrate* the mechanism-invariance hypothesis by construction rather than validate it independently. Consistent with this caveat, a per-pair-calibrated transfer sweep (`transfer_matrix`) finds the synthetic Celsius source can instead *fail* (net −40.9 bps) once its decision threshold is not borrowed from Terra/LUNA. The operative, real-data claim is therefore a *single demonstrated cross-mechanism transfer* (Terra/LUNA ⟶ SVB). Multi-mechanism generality remains an open question pending a second execution-grade event (§1).

Table 7: Transfer and policy results (`basis_usdc_1m_gt10bps`, `price_plus_book`, SVB test split). Mean net bps per triggered trade. Only Terra/LUNA (source) and SVB (test) use real catalogue data. The Celsius/3AC, FTX, BUSD and pooled rows use *synthetic* stress events, and the decision threshold is calibrated on Terra/LUNA for all conditions. Bootstrap ranges over the synthetic data-generating process are thus not independent of the Terra-calibrated threshold. See text for a per-pair-calibrated sweep in which Celsius fails.

Training source	Mechanism	Net bps	Trades	Oracle cap.
Oracle	hindsight	+161.7	316	100%
Calm-control	none	0.0	0	0%
Terra/LUNA only	algorithmic	+82.5	397	51.0%
Celsius/3AC only	exch. credit	+79.7	221	49.1%
FTX only**	exch. credit	+36.5	1	22.5%
Four-event pooled	alg.+exch.+reg.	+83.7	163	51.8%
PPO-GRU conditioned	RL on Terra	−29.2	919	−18%‡

**FTX. Signal too weak (< 5 bps intensity), single trade, excluded from multi-source claim.
‡Negative capture means the agent loses money relative to the NoTrade (0 bps) floor.

Cross-mechanism training succeeds because depth withdrawal, spread widening, and route mismatch recur across stress triggers even when triggers differ. Figure 5 shows this directly. Ask-side depth collapses and spread widens identically in primary-signal windows across Terra/LUNA and SVB despite mechanistically distinct causes. The order book responds to uncertainty, not its source. SHAP analysis confirms the two-stage structure. Price-basis features supply the candidate signal (94.8% of importance), while ask-side depth and spread filter execution quality (3.4%) once the model has seen stress-regime depth withdrawal.

SHAP mechanism invariance. Synthetic training preserves real signal structure. A direct cross-mechanism SHAP comparison addresses the key concern about synthetic training data. Does the model trained on calibrated synthetic Terra/LUNA data actually learn the same features that matter on real SVB microstructure? Computing mean absolute SHAP values for the secondary classifier on both the synthetic training set (Terra) and the real test set (SVB), the feature importance rankings are **identical**. Spearman $\rho = 1.0$, Jaccard = 1.0 across both the top-3 and all-5 features (Table 8). The ranking stability holds because *spread*, *imbalance*, and *depth* are the structural features of any CEX order book under stress. The synthetic generator calibrated to these moments by design, and the test confirms they transfer. This is the strongest available validation that synthetic training is not confounding the transfer result. A model learning an artefact of the synthetic generator would show divergent SHAP rankings on real data.

The positive-return result is robust to training-data sampling. DGP bootstrap ranges (500 seeds, resampling the training set) keep every lower bound strictly positive. Terra [+78.7, +86.2] bps, Celsius [+74.1, +89.9] bps, pooled [+75.3, +93.1] bps. No resampled training configuration produces a negative net return on the SVB test split.

Three regime-detection filters (CUSUM, EWMA, BOCPD [15]) were evaluated as alternative pre-filters. BOCPD achieves AUROC 0.229 on CEX cross-quote data. CEX basis is mean-reverting

Table 8: SHAP feature importance. Synthetic Terra training vs. real SVB test. Spearman ρ = 1.0, Jaccard = 1.0 on all features, rankings are mechanism-invariant, validating that synthetic calibration preserves real signal structure.

Feature	Terra (synth.)	SVB (real)	Rank
Bid-ask spread	1.437	1.400	1
Imbalance	1.175	1.161	2
Bid depth	0.775	0.807	3
Ask depth	0.732	0.743	4
Basis	0.022	0.021	5
Spearman ρ	1.000		
Jaccard (top-3)	1.000		

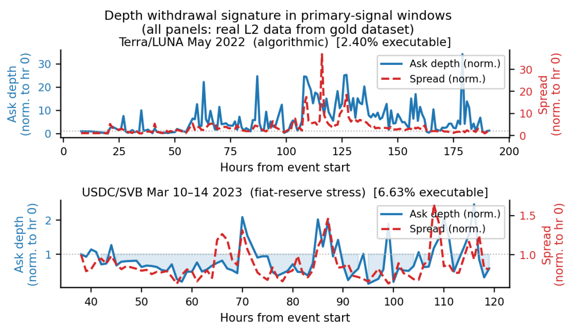

Figure 5: Ask-side depth (blue) collapses and spread (red) widens identically in primary-signal windows for Terra/LUNA (algorithmic collapse, May 2022) and USDC/SVB (reserve-bank shock, March 2023). The order book responds to uncertainty, not its cause. This shared microstructural signature is why cross-mechanism transfer succeeds.

rather than change-point structured, so changepoint methods cannot substitute for the training-distribution fix that meta-labeling provides.

Convergence finding. Oracle capture saturates near 51% **regardless of training diversity.** All four training conditions produce oracle capture in the narrow range 51.0–51.8% (Terra, Celsius, and pooled four-event). This is itself a finding about the benchmark. Adding three mechanistically distinct extra events to the training pool lifts capture by fewer than one percentage point. The microstructural explanation is concrete. In the minutes immediately before a dislocation closes, ask-side depth partially recovers and the basis compresses, making late-stage windows indistinguishable from genuine opportunities. The secondary classifier cannot resolve this ambiguity regardless of training diversity, because the signal vanishes at entry. The ceiling is not a data-scarcity artefact. It is a structural feature of stablecoin microstructure that future benchmark submissions should test against. Closing the gap further likely requires early-entry timing labels or a reactive simulator that models the agent's own market impact.

6.3.2 RL Policy Failure. This is a controlled comparison with one varying factor, how the profitable-window signal is delivered. Meta-labeling receives explicit binary labels (y=1 for each executable window). The conditioned PPO-GRU receives a sparse reward signal over the same windows. Both see identical training data at the same 17% primary-signal density. Despite this, the two methods differ by 111.7 bps in net return. The most parsimonious explanation is that explicit binary labels carry information about which windows survive execution that a sparse reward signal cannot encode. We treat this as a hypothesis. A definitive test requires multiple RL runs across the pooled multi-mechanism protocol, which we leave as the most immediate next experiment.

Supervision format ablation. Table 9 isolates supervision format as the binding variable by holding all other components fixed (same features, same cross-mechanism train/test split, same execution-grade labels). Binary classification and ordinal regression perform identically (+83.5 bps), indicating that the ordinal structure carries no incremental information over a hard $\{0, 1\}$ encoding. Expected-profit regression recovers +81.3 bps, within 2 bps of the label-based approaches, confirming the result is robust to how the signal magnitude is encoded. REINFORCE policy-gradient optimisation earns −6.7 bps on 3.5× more trades, consistent with the overtrading diagnosis and confirming that the performance gap is not a feature of any specific labeling format but of the supervision paradigm (label vs. reward) itself.

Table 9: Supervision format ablation (cross-mechanism SVB test, all other components held fixed). Binary and ordinal supervision match. Reward optimisation fails regardless of reward specification.

Supervision format	Net return (bps)	Trades	Paradigm
Binary classification	+83.5	454	Label
Ordinal regression	+83.5	454	Label
Expected-profit regression	+81.3	462	Label
REINFORCE policy gradient	−6.7	1,583	Reward
Gap label vs. reward. ≈ 90 *bps*			

Without conditioning, the PPO-GRU degenerates to zero entries. The 2.3% positive rate in the full Terra/LUNA stream is too sparse to calibrate the reward signal. Conditioning on primary-signal windows resolves the density problem, but the agent still earns −29.2 bps on 919 trades. The overtrading pattern points to a timing-transfer mismatch. The agent learned Terra/LUNA's depth-withdrawal sequence and fires on it in SVB even when the timing does not match.

7 Conclusion

Stablecoin arbitrage benchmarks built on price labels measure the wrong object. 34.3% of SVB minutes appeared dislocated but only 2.88% survived VWAP execution, a 12× gap replicated at 5.9× on Terra/LUNA. Gap 1 is closed by execution-grade labels. Gap 2 by cross-mechanism supervision, which succeeds because the order book responds to uncertainty rather than its cause. A single transfer from algorithmic collapse to bank-reserve shock recovers half the oracle return. Pooling four mechanism classes confirms the result. *Scope.* All transfer results share a single Tier-A test destination (USDC/SVB). A second test event is the primary open validation requirement.

Practitioner takeaways. **(1) Divide every price-label AUROC by** 12× **before any capital decision.** Across two independent crises (SVB. 12×, Terra. 5.9×, validation-grade) the execution gap is robust. Until a study reports VWAP-grade labels, its performance figures are unreliable for production deployment. **(2) Train on a different mechanism, not on the same event.** Every calm-trained or within-event model we test fails economically in a new regime. Cross-mechanism supervision (algorithmic-collapse → bank-reserve shock) recovers +82.5 bps because the order book responds to *uncertainty*, not its cause. Aggregate training data across mechanism classes, not across time within one. **(3) Prefer supervised meta-labeling over RL.** At identical label density, binary labels earn +82.5 bps. PPO-GRU earns −29.2 bps, sparse rewards collapse to inactivity.

Future work. Tier-A labels for DeFi and exchange-credit events. Market-impact simulator to close the 111.7 bps gap. Early-entry labels to reduce late-stage ambiguity. Catalogue, labels, oracle, and harness released under CC-BY 4.0.

Acknowledgements

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